LIFTING A FACTORIZATION THROUGH A SQUARE-ZERO IDEAL: ONE STEP WITH FINITELY GENERATED KERNEL

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Standard smooth refactoring of an ind-smooth map. lemma-colimit-standard-smooth (carded L26800). Gives the standard smooth presentation of the mod-I factorization.
- Smooth maps are formally smooth. Algebra, proposition smooth-formally-smooth (cross-chapter, named). Lifts the mod-I isomorphism B/IB -> A'/IA' to an R-algebra map B -> A'.
- Nakayama's lemma. Algebra, lemma NAK (cross-chapter, named). Promotes the mod-I surjection B -> A' to a surjection.
- Finite presentation of kernels. Algebra, lemma finite-presentation-independent (cross-chapter, named). The kernel J of B -> A' is finitely generated and contained in IB.

Source reference. Original reference 00DV: focused-seeds/geometry/S9-smo-lift-once/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00R2: focused-seeds/geometry/S9-smo-lift-once/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TN: focused-seeds/geometry/S9-smo-lift-once/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CI: sources/stacks-smoothing/pinned-original-source.tex, source lines 892–892. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that*

(1) $I^2 = 0$, *and*

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) $\Lambda/I\Lambda$ *is a filtered colimit of smooth* R/I*-algebras.*

Let $\varphi : A \to \Lambda$ *be an* R*-algebra map with* A *of finite presentation over* R*. Then there exists a factorization*

$$A \to B/J \to \Lambda$$

where B *is a smooth* R*-algebra and* $J \subset IB$ *is a finitely generated ideal.*

Proof. Choose a factorization

$$A/IA \to \bar{B} \to \Lambda/I\Lambda$$

with $\bar{B}$ standard smooth over R/I; this is possible by assumption and Lemma 07CI. Write

$$\bar{B} = A/IA[t_1, \ldots, t_r]/(\bar{g}_1, \ldots, \bar{g}_s)$$

and say $\bar{B} \to \Lambda/I\Lambda$ maps t_i to the class of λ_i modulo $I\Lambda$. Choose $g_1, \ldots, g_s \in A[t_1, \ldots, t_r]$ lifting $\bar{g}_1, \ldots, \bar{g}_s$. Write $\varphi(g_i)(\lambda_1, \ldots, \lambda_r) = \sum \epsilon_{ij}\mu_{ij}$ for some $\epsilon_{ij} \in I$ and $\mu_{ij} \in \Lambda$. Define

$$A' = A[t_1, \ldots, t_r, \delta_{i,j}]/(g_i - \sum \epsilon_{ij}\delta_{ij})$$

and consider the map

$$A' \longrightarrow \Lambda, \quad a \longmapsto \varphi(a), \quad t_i \longmapsto \lambda_i, \quad \delta_{ij} \longmapsto \mu_{ij}$$

We have

$$A'/IA' = A/IA[t_1, \ldots, t_r]/(\bar{g}_1, \ldots, \bar{g}_s)[\delta_{ij}] \cong \bar{B}[\delta_{ij}]$$

This is a standard smooth algebra over R/I as $\bar{B}$ is standard smooth. Choose a presentation $A'/IA' = R/I[x_1, \ldots, x_n]/(\bar{f}_1, \ldots, \bar{f}_c)$ with $\det(\partial \bar{f}_j/\partial x_i)_{i,j=1,\ldots,c}$ invertible in A'/IA'. Choose lifts $f_1, \ldots, f_c \in R[x_1, \ldots, x_n]$ of $\bar{f}_1, \ldots, \bar{f}_c$. Then

$$B = R[x_1, \ldots, x_n, x_{n+1}]/(f_1, \ldots, f_c, x_{n+1} \det(\partial f_j/\partial x_i)_{i,j=1,\ldots,c} - 1)$$

is smooth over R. Since smooth ring maps are formally smooth (Algebra, Proposition 00TN) there exists an R-algebra map $B \to A'$ which is an isomorphism modulo I. Then $B \to A'$ is surjective by Nakayama's lemma (Algebra, Lemma 00DV). Thus $A' = B/J$ with $J \subset IB$ finitely generated (see Algebra, Lemma 00R2). $\square$